

Translation of certificate Full Stack Web Developer

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<https://grellmann.app>

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Mr. Martin Grellmann, born on July 22, 1975, took part in the continuing education program "Individual training program Media" from January 6, 2020 to August 1, 2021 and obtained the qualification **Full Stack Web Developer**.

Certificate Full Stack Web Developer

Mr. Martin Grellmann, born on July 22, 1975, took part in the continuing education program "Individual training program Media" from January 1, 2020 to August 1, 2021 and achieved the results listed below.

Overall grade: 'sehr gut' – very good

Leipzig, August 1, 2021

Grading scale

Grade (number)	Grade (description)		Letter grade	GPA
1	'sehr gut' – very good	corresponds to	A	4.0
2	'gut' – good		A	3.0
3	'befriedigend' – satisfactory		B	2.0
4	'ausreichend' – sufficient		C	1.0
5	'mangelhaft' – poor		D	0.0

Modules

Module	Teaching units	Grade
Internship	32 TU	participated
Responsive web design with HTML5 and CSS3	160 TU	'sehr gut' – very good
Content management for programmers	160 TU	'sehr gut' – very good
Search engine marketing	40 TU	'sehr gut' – very good
Search engine optimization	120 TU	'sehr gut' – very good
Keyword advertising, Google Analytics	40 TU	'sehr gut' – very good
Social media marketing	120 TU	'sehr gut' – very good
Online PR, social media marketing, writing for the web	40 TU	'gut' – good
E-commerce, shop systems	160 TU	'sehr gut' – very good

German Version:

https://grellmann.app/assets/reference/de/zertifikat_full_stack_web_developer.pdf

Project management	80 TU	external certificate
Scripting languages: JavaScript	160 TU	'sehr gut' – very good
JavaScript development with jQuery and AJAX	160 TU	'sehr gut' – very good
Scripting languages: PHP	160 TU	'sehr gut' – very good
Databases, MySQL	160 TU	'sehr gut' – very good

The learning outcomes are assessed and evaluated by means of written examinations, written assignments, documentation and practical projects. The scope of the modules is specified in teaching units (TU) of 45 minutes each.

Leipzig, August 1, 2021
(signed)

Course content

Responsive web design

- Responsive web design: definition, basics, criteria
- HTML5, CSS3 and JavaScript in use for responsive web design
- Possibilities of mobile devices
- Screen resolutions, settings, browser settings and problems
- Operating systems and user requirements
- Usability factors
- Navigation in responsive web design
- Criteria for loading times in responsive web design
- Media data and responsive web design
- Mobile websites with jQuery Mobile and frameworks
- Accessibility and search engine optimization
- Putting a website online and server hosting

Content management systems: WordPress for programmers

- WordPress as a website content management system and as a blog
- Setting up and customizing local test servers
- Installation and maintenance of WordPress
- Editing, language conversion, basic settings
- Installation and activation of themes
- Settings and development of custom themes
- Page structures and menu management
- Categories, articles, comments, content
- Working with child themes
- Installing, using and customizing plugins
- Designing and creating custom plugins for WordPress
- Creating widgets
- Plugin recommendations
- Accessibility, search engine optimization, social media
- User and rights management
- WordPress system optimization
- Multilingualism in WordPress

Content management systems: TYPO3 for programmers

- TYPO3 basics, system requirements, areas of application
- TYPO3 as a website content management system
- Setting up and customizing local test servers
- Installation and configuration of TYPO3
- Editing, language conversion, basic settings
- Introduction to the TYPO3 backend and file overview
- Design and use of design templates in TYPO3
- Creating custom HTML templates: HTML5/CSS
- Templates and implementation in TYPO3

- Introduction to and use of TypoScript
- Page structures, page types, menu management, content
- TYPO3 and extension software
- Multilingualism: basics and structure
- Efficient use of extensions and customization
- Print output with TYPO3
- Accessibility and search engine optimization
- Use and application of backend layouts
- User and rights management, frontend/backend

Search engine optimization and marketing

- Introduction to search engines
- Search engine optimization (SEO): on-page and off-page optimization
- Keyword strategies
- Monitoring and controlling tools
- Search engine advertising (SEA): Google AdWords campaigns
- Google Analytics
- Project work

Social media marketing

- Web 2.0 / social media marketing overview
- Social media marketing strategies
- Social media services, social networks
- Social media monitoring
- Social media analysis
- Conception of social media campaigns
- Project work

Online PR (for online marketing managers)

- Online journalism
- Introduction to online marketing
- Email marketing
- Writing for the web

E-commerce and shop systems: WooCommerce

- Setup of a local test server
- Installation of WooCommerce and boilerplate
- Installation, user interface and basic settings
- Templates, extensions and language changeover
- Category and article management, article types
- Attribute assignment
- Manufacturer and customer management
- Shipping options, payment systems, tax rules
- Invoice processes and shopping cart
- Cross-selling and up-selling methods
- Customer loyalty methods (email, newsletter, reviews)

- Legal requirements for shop systems
- User and rights management
- Accessibility and search engine optimization

Project management specialization

- Including external certificate

Scripting languages: JavaScript

- Structure and introduction to JavaScript
- JavaScript and HTML5 in interaction
- Language elements in JavaScript
- Use of variables, functions, methods and parameters
- Use of control structures (conditions and loops)
- Form evaluation and form control
- JavaScript and extensions (jQuery and AJAX)
- JavaScript and OOP: use and structure
- JavaScript in practice, fallbacks

JavaScript development with jQuery and AJAX

- Overview of Web 2.0, AJAX and HTML5
- JavaScript and AJAX in interaction
- AJAX basic structure and basic components
- Elements and attributes / XML syntax
- XMLHttpRequest object and data queries
- Integration of jQuery libraries
- jQuery basic structure and basic components
- Event handling and structure of jQuery
- Animations and effects
- RIA (Rich Internet Applications)
- jQuery and mobile devices
- Overview of alternative libraries
- Use and structure of Express.js
- Use and structure of Node.js

Scripting languages: PHP

- Setting up local test servers
- Structure and introduction to PHP, language elements in PHP
- Use of variables, functions, methods and parameters
- Use of control structures (conditions and loops)
- PHP and forms
- Creating data storage with PHP
- Dynamic visualization of content, data and folders
- PHP and HTML5 in interaction
- Working with data structures (data records, tables)
- Creating, changing, querying and deleting data records
- PHP and PDO with SQL

- PHP and OOP: use and structure
- Server hosting and data management

Databases and SQL (MySQL)

- Setting up local test servers
- Databases and database systems
- Database structure and relational databases
- Entity relationship model (ERM)
- MVC handling and structure with PHP and SQL
- Primary and foreign keys
- Normalization
- Referential integrity in databases
- Overview of the language structure in SQL, SQL standard functions
- Interaction of PHP and SQL (MySQL)
- Key fields, indexing, queries
- Viewing data entries
- Transaction management
- Managing access rights and users in databases